

Elastography in the Child's Liver.

In the last two chapters we looked at pictures of the liver. Ultrasound, CT and MRI show us the size of the liver, its texture, its vessels and its bile ducts. This chapter is about something different. It is about a single number: how stiff the liver is.

Why does stiffness matter? Because scar tissue is stiff. A healthy liver is soft, a little like a ripe peach. As fibrosis builds up, the liver becomes firmer, and a cirrhotic liver is hard. For many years the only way to know how much scar there was in a child's liver was to take a biopsy.

Elastography promises to give us an estimate of fibrosis without a needle, in a few minutes, and as often as we like.

That promise is real, and in adults elastography has replaced many biopsies. But children are not small adults. Their normal values are different, they cannot hold their breath or lie still on command, and many of the diseases we see in children make the liver stiff for reasons that have nothing to do with scar.

The main idea of this chapter is simple. A stiff liver is not always a scarred liver. And the number you get depends on the machine, the probe and the child, as much as on the fibrosis. If you remember those two sentences, you will avoid most of the mistakes that are made with elastography in children.

So this chapter asks seven questions. First, how do the different machines measure stiffness? Second, what makes a reading valid in a child? Third, what is a normal value in a child? Fourth, what can make a liver stiff without fibrosis? Fifth, can we use the adult cut-offs, and what do the guidelines say? Sixth, how well does elastography work in the diseases we see most often, such as biliary atresia, cystic fibrosis, Fontan circulation and fatty liver? And seventh, what about measuring fat with ultrasound, and what does MR elastography add?

A short note on the sources. The most detailed document is the 2024 guideline of the World Federation for Ultrasound in Medicine and Biology, usually called WFUMB, written by Ferraioli and colleagues. It builds on the

2020 consensus of the Society of Radiologists in Ultrasound, written by Barr and colleagues. The American liver society, AASLD, published a guidance on imaging-based non-invasive liver disease assessment, written by Sterling and colleagues and printed in 2025. For portal hypertension in adults we use the Baveno seven consensus. For fatty liver in children we use the AASLD paediatric guidance of 2025, written by Xanthakos and colleagues, and the ESPGHAN position paper of 2026, written by Mann and colleagues. As in the previous chapters, most of the paediatric numbers come from single studies rather than from guidelines, and we will name them as we go.

A note on words. Sensitivity is the proportion of children with the disease whom the test picks up. Specificity is the proportion of children without the disease whom the test correctly calls normal. The positive predictive value is the chance that a child with a positive test really has the disease. The negative predictive value is the chance that a child with a negative test really does not. The area under the curve, or AUROC, is a single number between one half and one that describes how well a test separates two groups. One half is no better than tossing a coin. One is perfect. Anything above about zero point eight is considered good.

Fibrosis stages are written with the letter F. F zero is no fibrosis, F one and F two are mild and moderate fibrosis, F three is advanced, or bridging, fibrosis, and F four is cirrhosis.

Part one. How the machines work.

Every kind of elastography works on the same idea. We send a gentle push into the liver. The push creates a small wave that travels sideways through the tissue. This is called a shear wave. Shear waves travel faster in stiff tissue and slower in soft tissue. So if we measure the speed of the shear wave, we know how stiff the liver is.

The speed is measured in metres per second. Most machines then convert the speed into a stiffness, measured in kilopascals. When you read a report, check whether the number is a speed or a stiffness, because one point five metres per second and one point five kilopascals mean very different things.

There are four main methods.

The first is vibration-controlled transient elastography, usually called by its brand name, FibroScan. Here the push is mechanical. The tip of the probe gives a small punch to the skin, and the machine follows the wave as it passes through the liver. There is no ordinary ultrasound picture on the screen, so the operator cannot see exactly where the measurement is being taken. The machine samples a cylinder of liver about one centimetre wide and four centimetres long, at a depth of twenty-five to sixty-five millimetres with the standard M probe. The result is given in kilopascals.

In children the choice of probe matters. The probe is chosen by the size of the chest. The small S one probe is for a chest circumference below forty-five centimetres. The S two probe is for a chest between forty-five and seventy-five centimetres. The M probe is for a chest above seventy-five centimetres, and the XL probe is for obese patients.

The second method is point shear wave elastography, sometimes called ARFI, which stands for acoustic radiation force impulse. Here the push is made by a focused pulse of sound, sent from an ordinary ultrasound probe. The operator sees the liver on the screen and places a small box, the region of interest, about fifteen to twenty millimetres below the liver capsule. The machine measures stiffness at that one point. It needs a short breath-hold, and each vendor has its own scale, with few normal values for children.

The third method is two-dimensional shear wave elastography, or two-D shear wave for short. Here the machine makes many pushes in a row and builds a colour map of stiffness over a larger area. The operator chooses a region of at least ten millimetres on the live image and takes three to five readings. This method works even in newborns, and it can be done in free breathing using a long video clip. Again, the numbers are specific to each vendor.

The fourth method is MR elastography. A small passive driver, like a flat drum, is placed on the abdomen and vibrates at sixty hertz. The MRI scanner then images the waves across large slices of the whole liver. MR elastography samples much more liver than any ultrasound method. But it needs a scanner and a still child, which usually means sedation below school age.

There is one important trap here. Ultrasound machines report a quantity called Young's modulus. MR elastography reports a different quantity, the shear stiffness. For the liver, the shear stiffness is about one third of Young's modulus. So a value of three kilopascals on MR elastography is roughly the same liver as nine kilopascals on ultrasound. Never compare an MR elastography number with an ultrasound number.

Finally, the FibroScan machine can also measure fat. As the ultrasound signal passes through fatty liver, it loses more energy. This loss is called attenuation, and the FibroScan reports it as the controlled attenuation parameter, or CAP, measured in decibels per metre. We will come back to CAP in Part seven.

Part two. What makes a reading valid.

The rules for a good elastography examination were written for adults. In children some of these rules bend, and some of them break.

The first rule is fasting. Adults are asked to fast for four hours and to rest for ten minutes before the scan. This matters, because after a meal more blood flows into the liver and the liver becomes stiffer for up to about an hour. In one study, more than a third of normal livers rose above six kilopascals after breakfast. But an infant cannot fast for four hours. The practical answer is to scan the baby just before the next feed, when the stomach is as empty as it will get.

The second rule is the breath-hold. Adults hold their breath in a relaxed, neutral position. Young children cannot do this, so they are scanned in free breathing. A Korean study by Yoon and colleagues showed that free breathing gives values about eleven percent lower than a breath-hold on two-D shear wave. That is not a big difference, but it means that values taken in different ways should not be compared directly.

The third rule is the quality of the FibroScan examination. We need at least ten valid measurements, and the spread of the measurements, called the interquartile range divided by the median, should be thirty percent or less. This rule is the same in children. But invalid examinations are much more common in younger children. In one paediatric series, children with failed

examinations had a mean age of about six years, compared with about ten years for children with good ones.

The fourth rule is no sedation. In adults this rule rarely comes up. In children it does, because many young children are sedated for imaging. In the large paediatric meta-analysis by Li and colleagues, sedation on its own raised the measured stiffness.

The fifth rule is the choice of probe. As we said, children need the small S probes. But in the same meta-analysis, use of the S probe was also linked to higher values. So a value from an S probe is not exactly the same as a value from an M probe.

The sixth rule concerns change over time. The WFUMB guideline says that, on the same machine, a change of more than ten percent is probably real. In children, however, two different operators measuring the same child can differ by up to about thirty-two percent either way. So in a child, a small change from one visit to the next may be nothing more than the difference between operators.

Part three. What is normal in a child.

In adults, a normal liver reads below about five kilopascals on FibroScan. In children the normal value is lower in infancy and rises with age.

Two studies give us a good picture. The first comes from Heidelberg, published by Engelmann and colleagues in 2012. The median value was four point four kilopascals at nought to five years, four point seven at six to eleven years, and five point one at twelve to eighteen years. The upper limit of normal rose from six point nought to six point eight kilopascals across the same age groups.

The second comes from Osaka, published by Tokuhara and colleagues in 2016. Their values were lower. The median was three point four kilopascals in the youngest group, three point eight in the middle group and four point one in teenagers. The ninety-fifth centile rose from four point six to seven point nine kilopascals.

The best single summary comes from the individual patient data meta-analysis by Li and colleagues, published in 2020. They pooled ten studies

of six hundred and fifty-two healthy children. The normal range for FibroScan was two point four five to five point five six kilopascals. Four things pushed the value up: older age, sedation, use of the S probe, and fat in the liver on ultrasound.

So a useful rule is this: a normal FibroScan in a child is below about five and a half kilopascals.

But normal also depends on the machine. In a Norwegian study by Mjelle and colleagues, the same healthy children aged four to seventeen were scanned on several machines. The median on two-D shear wave was three point three kilopascals, while point shear wave and FibroScan both gave about four point one. The same liver, measured on the same day, gave a different number on each platform.

In newborns, a French study by Charlier and colleagues using one two-D shear wave machine found a median of six point nought kilopascals at birth. This fell to about five point three by adolescence. So on this machine, the newborn liver reads stiffer, not softer.

MR elastography has its own normal values. In healthy children aged eight to seventeen, Sawh and colleagues found a mean of two point four five kilopascals, with a ninety-fifth centile of three point one nine. This is higher than in healthy adults, who average about two point one. Worse, up to about a quarter of carefully screened healthy children had a value that would suggest fibrosis by published thresholds.

The lesson from this part is simple. A paediatric elastography report must always name the method, the probe and the machine. Without them, the number cannot be interpreted.

Part four. When stiff is not scarred.

Elastography measures stiffness, not scar. Anything that makes the liver stiffer will raise the number. In children, several of these things are common.

The first is inflammation. In acute hepatitis the liver is swollen with inflammatory cells and fluid. Arena and colleagues studied eighteen adults with acute viral hepatitis. At the peak of the ALT, every single one of them

had a stiffness above the cut-off for cirrhosis. As the hepatitis settled, the values came down. In children, think of autoimmune hepatitis at presentation, acute Wilson disease, viral hepatitis and rejection after transplant.

A study from Boston by Raizner and colleagues looked at one hundred and fifty-four children and young adults who had a FibroScan within a year of a liver biopsy. In those with no or mild fibrosis, stages F zero to F two, a high ALT pushed the stiffness above eight point six kilopascals, which is in the range we would call advanced fibrosis. In those who already had advanced fibrosis, the ALT made no difference. In other words, inflammation inflates the low end, which is exactly where most children sit.

The practical rule is this. In a child with mild disease, repeat the measurement once the ALT is below twice the upper limit of normal before you call it fibrosis. A very high value in a child who is known to be cirrhotic is still cirrhosis. But a high value in a child with an ALT of six hundred is not yet anything.

The second is cholestasis caused by blocked bile ducts. When bile cannot drain, the liver swells with bile and becomes stiff. Millonig and colleagues showed this in two ways. In pigs, tying the bile duct raised the stiffness from four point six to eight point eight kilopascals within two hours. In patients with tumours blocking the bile duct, the stiffness fell from fifteen point two to seven point one kilopascals within three to twelve days after the duct was drained. In children, this matters in biliary atresia, choledochal malformations and gallstones.

The third is congestion. When the pressure in the veins draining the liver rises, the liver fills with blood and becomes stiff. In the same research group's pig experiment, clamping the inferior vena cava raised the stiffness from three point one to twenty-seven point eight kilopascals, and it went back to normal within five minutes of releasing the clamp. In adults with decompensated heart failure, the stiffness was about forty-one kilopascals, and it fell to about eighteen once the heart failure was treated. In children with a Fontan circulation, at a median age of about fourteen years, the stiffness was eighteen point six kilopascals, compared with four point seven in healthy children of the same age. In children, think of Fontan,

right heart failure, restrictive cardiomyopathy and sinusoidal obstruction syndrome.

The fourth is food, as we discussed in Part two. Infants fed on demand are never truly fasted.

The fifth is infiltration. When the liver is filled with abnormal cells or material, such as amyloid, tumour cells or storage cells, it becomes stiff. In children, think of haemophagocytic lymphohistiocytosis, storage diseases and leukaemia.

The sixth is fat. Moderate or severe steatosis makes FibroScan overestimate fibrosis. This matters in teenagers with fatty liver.

Finally, ascites and obesity. These usually cause the examination to fail rather than to give a false value. FibroScan is the most affected and MR elastography the least.

Notice one thing. In every example we have just given, the value before the cause was removed was above fifteen kilopascals. Fifteen kilopascals is the adult cut-off that rules in advanced chronic liver disease. So inflammation, cholestasis and congestion can each push a liver with no cirrhosis into the cirrhotic range. The rule is: measure after drainage, after decongestion, or after the ALT has settled. Otherwise, do not interpret the number.

Part five. Adult cut-offs and what the guidelines say.

In adults there are two simple rules that every fellow should know. They are called the rule of five and the rule of four.

The rule of five comes from the Baveno seven consensus and applies to FibroScan. Below five kilopascals is normal. Below ten kilopascals rules out compensated advanced chronic liver disease. Between ten and fifteen is suggestive and needs confirmation. Above fifteen rules it in. Between fifteen and twenty kilopascals, with platelets below one hundred and ten, suggests clinically significant portal hypertension. Above twenty-five kilopascals rules it in. And an adult with a stiffness below twenty kilopascals and platelets of one hundred and fifty or more can safely skip screening endoscopy for varices.

The rule of four comes from the Society of Radiologists in Ultrasound and applies to the acoustic push methods. Five kilopascals or less is normal. Below nine rules out advanced disease, nine to thirteen is suggestive, and above thirteen rules it in. Above seventeen suggests portal hypertension, and above twenty-one rules it in.

Can we use these rules in children? The WFUMB guideline says clearly that the rule of four cannot be applied to children. Only the normal value of about five kilopascals fits children well. For the rule-out and rule-in values, children have different diseases and different confounders, and for portal hypertension there are almost no paediatric data. One study from Taiwan, by Ling and colleagues in 2024, applied the Baveno seven rule for skipping endoscopy to forty-eight children with biliary atresia, and it missed no varices that needed treatment. That is encouraging, but it is one small study.

What do the two main imaging guidelines say about children?

The AASLD guidance says that thresholds in children differ from adults and depend on the disease, and it does not endorse any. It also says there is not enough evidence to prefer one method over another.

The WFUMB guideline says that no paediatric cut-offs can be recommended. But it then offers a practical rule: five kilopascals or less rules out significant disease, and fifteen kilopascals or more rules it in, except in biliary atresia, cystic fibrosis liver disease and Fontan circulation. It also says that shear wave methods are helpful for diagnosis and follow-up in children.

Both guidelines list the same confounders: congestion, blocked bile ducts and inflammation, and AASLD adds oedema, fasting and sedation.

They differ on follow-up. AASLD says elastography should not be used on its own to judge whether fibrosis is getting better or worse. WFUMB says that, on the same machine and probe, a change of more than ten percent is meaningful.

They also differ on biliary atresia. AASLD says that severe cholestasis makes elastography less accurate. WFUMB says that elastography is useful to separate biliary atresia from other causes of infant cholestasis.

On Fontan they agree. Congestion raises the stiffness regardless of fibrosis, there are no thresholds, and the trend from the child's own baseline is what matters.

Part six. Disease by disease.

Let us now go through the diseases where elastography is used most in children.

We begin with biliary atresia before the Kasai operation. Here elastography is used to help decide which cholestatic infant is most likely to have atresia. A meta-analysis by Wagner and colleagues in 2022 pooled eleven studies of one thousand three hundred and seven infants. The pooled sensitivity for atresia was eighty-three percent and the specificity seventy-nine percent. In a prospective study of forty-one infants by Dillman and colleagues, a two-D shear wave speed above one point eight four metres per second was ninety-two percent sensitive and seventy-nine percent specific.

Elastography can also estimate fibrosis before the operation. In two hundred and sixteen infants studied by Chen and colleagues, a two-D shear wave value of thirteen kilopascals was the cut-off for advanced fibrosis, with a negative predictive value of ninety-six percent. In thirty-three infants studied by Yoon and colleagues, a cut-off of seventeen point five kilopascals detected advanced fibrosis or cirrhosis with a sensitivity of sixty-four percent and a specificity of eighty-six percent.

What does this mean in practice? About one in five infants who do not have atresia will still read as stiff, because cholestasis itself raises the number. So elastography ranks the risk. It can move an infant up the queue. It must never be used to delay the liver biopsy or the cholangiogram, which remain the tests that decide.

After the Kasai operation, the question changes. Now we want to know whether the child has developed cirrhosis and varices. For cirrhosis, the Wagner meta-analysis of nine studies found a sensitivity of ninety-six percent and a specificity of eighty-nine percent. This is one of the best results for elastography anywhere in paediatric hepatology.

For varices, the liver is useful but the spleen is better. In a Thai study by Chongsrisawat and colleagues of seventy-three children, a liver stiffness of twelve point seven kilopascals or more detected varices with a sensitivity of eighty-four percent and a specificity of seventy-seven percent. In a second Thai study by Sintusek and colleagues of fifty-one children, a liver stiffness of twelve point five kilopascals or more had a sensitivity of eighty percent and a specificity of seventy percent. But in the same children, a spleen stiffness of twenty-eight point nine kilopascals or more had a specificity of eighty-seven percent. Across nine studies, the pooled values for liver stiffness were a sensitivity of eighty-five percent and a specificity of seventy-six percent.

Two more studies support the spleen. In one hundred and fifty children from India, a spleen stiffness of twenty-seven point six kilopascals or more predicted varices that needed treatment. And in the Hannover series by Goldschmidt and colleagues, no child bled when the spleen stiffness was below sixty kilopascals. So after Kasai, measure both the liver and the spleen every year, and refer for endoscopy when the trend is rising or the spleen is above about twenty-eight kilopascals.

Next, cystic fibrosis. Liver disease in cystic fibrosis is patchy, so a single biopsy can miss it. The best study is by Lewindon and colleagues in 2019, who studied one hundred and sixty children with two biopsy passes each. Liver stiffness correlated well with the fibrosis stage. A value of five point five kilopascals flagged cystic fibrosis liver disease with an AUROC of zero point eight two. A value of eight point seven kilopascals identified advanced fibrosis or cirrhosis with a specificity of one hundred percent. When the stiffness was combined with the APRI score, a simple blood test score based on AST and platelets, the positive predictive value reached ninety-two percent. So in cystic fibrosis, use FibroScan together with APRI, not stiffness alone.

After liver transplantation, elastography is less helpful. In ninety-four children studied in Boston by Lee and colleagues, a value of six point five kilopascals detected moderate fibrosis or more with an AUROC of only zero point seven one. A low value is mildly reassuring, but it should not replace a protocol biopsy, and it should always be read together with the ALT and a Doppler ultrasound.

In the Fontan circulation, the liver is both congested and scarred. Fibrosis is almost universal by adolescence. The FibroScan value is very high, eighteen point six kilopascals compared with four point seven in controls, but most of this is congestion. In two hundred and six Fontan patients studied with MR elastography by Razavi and colleagues, the mean was four point four kilopascals, again very high for MR elastography. There are no thresholds. What matters is the trend. Use the same machine at every visit and compare the child with their own baseline. Never stage fibrosis from a single value.

In treated autoimmune hepatitis, primary sclerosing cholangitis and storage diseases, once there is no blockage and no active inflammation, the WFUMB guideline suggests that a value of about eight kilopascals points to advanced disease. This is based mostly on adult and mixed data, with a weak grade of recommendation. Use it as a trend, not as a firm threshold.

Finally, fatty liver, now called MASLD, which stands for metabolic dysfunction-associated steatotic liver disease. This is where the evidence is most surprising. A meta-analysis by Dong and colleagues in 2022, of seven studies and four hundred and thirty-six children, found that FibroScan was excellent at detecting advanced fibrosis. The sensitivity was eighty-nine percent, the specificity ninety-three percent, and the AUROC zero point nine six.

But in 2026, Nichols and colleagues published a prospective study from the American NASH Clinical Research Network. They studied ninety-two children at several centres, all with a biopsy within six months of the scan. Here the sensitivity was sixty-seven percent, the specificity seventy-six percent, and the AUROC only zero point six seven. The positive predictive value was forty percent, and the negative predictive value ninety percent.

Why the difference? Small, single-centre studies tend to look better than large, careful, multicentre ones. The prospective study is the one to believe. So in a teenager with fatty liver, FibroScan is a rule-out test at best. A low value is reassuring. A high value is right less than half the time.

The guidelines reflect this uncertainty. ESPGHAN 2026 says that a value below five point five kilopascals excludes advanced fibrosis and a value

above ten suggests it. AASLD 2025 says that adult thresholds have not been validated in children.

Part seven. Fat and MR elastography.

The FibroScan machine also reports CAP, the controlled attenuation parameter, which estimates fat in the liver.

CAP can separate a liver with fat from a liver without fat. In a study by Desai and colleagues of sixty-nine children with a biopsy, the mean CAP was one hundred and ninety-eight decibels per metre in children without steatosis and two hundred and ninety in children with steatosis.

But CAP cannot tell us how much fat there is. In severely obese children studied by Runge and colleagues, CAP was no better than a careful ordinary ultrasound, with AUROCs of zero point eight and zero point six eight, a difference that was not statistically significant. And in the NASH Clinical Research Network study, the median CAP was flat, between about three hundred and ten and three hundred and twenty-five, across all three grades of steatosis. In healthy children aged one to eighteen, the median CAP was about one hundred and eighty-three, and it did not change with age.

Here the guidelines disagree. AASLD 2025 says that CAP should not be used to diagnose fatty liver in children. ESPGHAN 2026 says CAP can be used as an extra test, with a cut-off above two hundred and eighty decibels per metre. The adult AASLD threshold is two hundred and seventy-five. For now, the only fat measurement in children that has been graded against biopsy is the MRI proton density fat fraction, which we met in the last chapter.

Now MR elastography. It is the most accurate method, and the hardest to get. It measures the stiffness of the whole liver, not a small sample. In ninety children with fatty liver studied by Schwimmer and colleagues, MR elastography identified advanced fibrosis with an accuracy of about eighty-nine percent. But for any fibrosis, the accuracy was only about seventy-two percent. Like ultrasound, it struggles at the early stages.

MR elastography has its own failure. When the liver is overloaded with iron, the MRI signal is lost and the standard sequence fails. This happens when the iron measurement, called R two star, is above about seventy-six per second on a one point five tesla scanner.

When should you ask for MR elastography? When the child already needs an MRI for another reason, or when ultrasound elastography fails, for example in severe obesity, in ascites, or when the space between the ribs is too narrow. And remember once more: MR elastography kilopascals and ultrasound kilopascals are on different scales.

Part eight. North America and Europe compared.

Let us bring the guidelines together.

On paediatric fibrosis thresholds, AASLD 2024 offers none and does not prefer any method. WFUMB 2024 also says that none can be recommended, but then offers the practical rule of five or less to rule out and fifteen or more to rule in. So Europe adds something that North America does not.

On fatty liver in children, AASLD 2025 says that adult FibroScan thresholds are not validated. ESPGHAN 2026 gives thresholds of below five point five to exclude and above ten to suggest advanced fibrosis. This is a material difference.

On CAP, AASLD 2025 says it is not a diagnostic tool. ESPGHAN 2026 calls it an adjunct with a cut-off above two hundred and eighty. They differ.

On MR elastography, both agree that it is useful and the most validated method for advanced fibrosis, but cut-offs in children still need validation.

On confounders, both agree: congestion, cholestasis, inflammation and food, with AASLD adding oedema and sedation.

On repeated measurements, they differ. AASLD says elastography alone should not be used to judge change. WFUMB says that a change of more than ten percent on the same machine is meaningful.

And on varices in children, there are no data from either side. Baveno seven is an adult rule, and there is no paediatric version.

Putting it into practice.

Let us finish with eight common situations.

A cholestatic infant who might have biliary atresia. Do a two-D shear wave at the same time as the fasting ultrasound. A high value raises the urgency. Never delay the cholangiogram because of a low value.

A child presenting with autoimmune hepatitis or acute hepatitis, with an ALT in the hundreds. Wait until the ALT is below twice the upper limit of normal, and then stage the fibrosis. Do not call cirrhosis at presentation.

A child with biliary atresia after the Kasai operation. Do a FibroScan of the liver and a spleen stiffness every year. A rising trend, or a spleen above about twenty-eight kilopascals, should prompt an endoscopy. Do not rely on the adult rule of five alone.

A teenager with a Fontan circulation. Use the same machine at every visit and look at the change from the child's own baseline. Do not stage fibrosis from one value.

An obese teenager with a raised ALT. Do a FibroScan once the raised ALT has been confirmed. A value below five and a half kilopascals is reassuring. Do not grade the fat with CAP.

A child with cystic fibrosis. Do a FibroScan together with an APRI score. A value of about five and a half to six kilopascals or more, together with a raised APRI, makes cystic fibrosis liver disease likely. Do not rely on stiffness alone.

A child after liver transplantation. Do a FibroScan together with the ALT and a Doppler ultrasound. A low value is only weakly reassuring. Do not skip the protocol biopsy.

And a child in whom ultrasound fails, or who needs an MRI anyway. Do an MR elastography, and read it on its own scale. Do not compare it with ultrasound values.

Things to remember.

A normal FibroScan in a child is below about five and a half kilopascals. It rises with age, sedation, use of the small S probe, and fat in the liver.

Each machine has its own normal. In the same child, two-D shear wave reads lower than point shear wave and FibroScan.

Inflammation, blocked bile ducts and congestion can each push a liver without cirrhosis above fifteen kilopascals. Measure after the cause has been removed, or do not interpret.

In biliary atresia before Kasai, stiffness ranks the infant. The biopsy and the cholangiogram decide.

After Kasai, spleen stiffness predicts varices better than liver stiffness.

In Fontan, read the trend from the child's own baseline, never the absolute number.

In paediatric fatty liver, FibroScan is a rule-out test, with a negative predictive value of ninety percent and a positive predictive value of only forty percent. CAP cannot grade fat.

MR elastography kilopascals and ultrasound kilopascals are on different scales. Never compare them.

Questions to test yourself.

Question one. What is the upper limit of normal for FibroScan in healthy children, according to the meta-analysis by Li and colleagues, and what four factors raise the value? Answer: about five point five six kilopascals. Older age, sedation, use of the S probe, and fat in the liver on ultrasound.

Question two. Why can a value of three kilopascals on MR elastography not be compared with three kilopascals on FibroScan? Answer: because ultrasound reports Young's modulus, while MR elastography reports the shear stiffness, which is about one third of Young's modulus. Three kilopascals on MR elastography is roughly nine kilopascals on ultrasound.

Question three. Name three causes of a high liver stiffness without fibrosis that are common in children. Answer: inflammation, for example autoimmune hepatitis at onset; blocked bile ducts, for example biliary atresia; and congestion, for example the Fontan circulation. Food, infiltration and fat are others.

Question four. In the Boston study by Raizner and colleagues, how did ALT affect liver stiffness? Answer: in children with no or mild fibrosis, a high ALT pushed the stiffness above eight point six kilopascals, into the advanced range. In children who already had advanced fibrosis, ALT made no difference.

Question five. Why should elastography not be used to decide whether a cholestatic infant goes to theatre? Answer: because about one in five infants without atresia also read as stiff, since cholestasis itself raises the stiffness. Elastography ranks the risk, but the biopsy and the cholangiogram decide.

Question six. After the Kasai operation, which measurement predicts varices better, and what cut-off was found in the Thai study? Answer: spleen stiffness. A spleen stiffness of twenty-eight point nine kilopascals or more had a specificity of eighty-seven percent.

Question seven. Why did the meta-analysis and the NASH Clinical Research Network study disagree about FibroScan in paediatric fatty liver, and which one should you believe? Answer: the meta-analysis pooled small single-centre studies and found an AUROC of zero point nine six. The prospective multicentre study with biopsies within six months found only zero point six seven. The prospective study is more reliable, so FibroScan is a rule-out test at best.

Question eight. What do AASLD and ESPGHAN say about CAP in children? Answer: AASLD 2025 says CAP should not be used to diagnose fatty liver in children. ESPGHAN 2026 accepts it as an adjunct with a cut-off above two hundred and eighty decibels per metre.

End of chapter.